CompoundIQ: AI-Driven Drug Combination Design System

3-Page Project Proposal

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Project Overview

CompoundIQ is an AI system that designs 3-compound drug formulations by understanding mechanism descriptions, generating candidate molecules, predicting interactions between compounds, and recommending safe, synergistic combinations. The system addresses the complex challenge of multi-drug therapy design where pharmaceutical scientists need to create combinations that work together safely without dangerous interactions.

Core Innovation: Unlike existing systems that only predict properties of single molecules, CompoundIQ designs complete 3-compound formulations while checking all pairwise interactions (A-B, A-C, B-C) and three-way combined effects (A+B+C), filtering out lethal combinations and ranking candidates by safety and efficacy.

Input: Text description of desired therapeutic mechanism
Process: Generate 125,000 possible 3-compound combinations → Screen for safety → Predict interactions → Rank by efficacy
Output: Top 10 combinations with detailed analysis + Top 500-1000 as downloadable datasets

Example:
Input: "Reduce muscle spasms via calcium blocking and GABA enhancement"
Output: Compound A (calcium blocker) + Compound B (GABA agonist) + Compound C (bioavailability enhancer) with full interaction analysis showing no dangerous metabolic conflicts, 1.52× synergistic effect, and 9.4/10 safety score.

System Architecture

Module 1: Mechanism Interpreter

- **Input:** Natural language mechanism description
- **Model:** BioBERT fine-tuned on ChEMBL mechanism annotations and DrugBank
- **Output:** Mechanism embedding vector + identified targets
- **Purpose:** Converts "blocks calcium channels" into molecular target requirements

Module 2: Compound Generator

- **Input:** Mechanism embedding
- **Process:** Molecular VAE trained on ZINC drug-like molecules generates 50 candidates per position (A, B, C)
- **Output:** 50 × 50 × 50 = 125,000 possible 3-compound combinations
- **Constraint:** All generated molecules must pass chemical validity (Lipinski's Rule, RDKit sanitization)

Module 3: Hierarchical Interaction Predictor

Level 1 - Pre-screening (Fast filters):

- Basic drug-likeness checks
- Obvious toxicity flags
- Eliminates ~80% → Down to 25,000 combinations

Level 2 - Pairwise Interactions (GNN models):

- Check A-B interactions: CYP enzyme competition, receptor binding, toxic metabolites
- Check A-C interactions: Same checks
- Check B-C interactions: Same checks
- Eliminates ~60% → Down to 10,000 combinations

Level 3 - Three-Way Interactions (Combined GNN):

- Predict A+B+C combined effects
- Check emergent properties
- Model metabolite cascades (A's metabolite interacting with B)
- Check for additive vs. synergistic effects
- Eliminates ~50% → Down to 5,000 safe combinations

Module 4: Safety Filter (Hard Constraints)

Auto-reject if:

- Both compounds inhibit same CYP enzyme >80%
- Predicted toxic metabolite probability >0.7
- Both prolong QT interval (cardiac risk)
- Known lethal interaction in DrugBank DDI database

- Predicted lethality score >0.3

Module 5: Multi-Objective Ranking

Scoring function:

```
Score = 0.35 × mechanism_match +
        0.30 × (1 - interaction_risk) +
        0.20 × synergy_potential +
        0.10 × synthesizability +
        0.05 × cost_estimate
```

Output: Top 1,000 combinations ranked by composite score

Module 6: Web Interface

- **Page 1:** Input mechanism → Generate combinations (20 min processing)
- **Page 2:** View top 10 with full detailed analysis
- **Page 3:** Browse top 500 in sortable/filterable table
- **Page 4:** Download top 1,000 as CSV/Excel/SDF

Datasets

Database	Size	Purpose	Access
ChEMBL	2.4M compounds	Drug structures + mechanisms	Free API
DrugBank	14K drugs, 140K interactions	Known drug-drug interactions	Free academic
TWOSIDES	1M drug combinations	Adverse event data	Free download
ZINC	230M compounds	Chemical space for VAE training	Free subsets
SIDER	1,430 drugs	Side effect profiles	Free download
STITCH	500K interactions	Protein-chemical interactions	Free download
PubMed	Millions of abstracts	Mechanism-structure relationships	Free API

Total training data: ~50,000 curated molecule-mechanism pairs from ChEMBL, 140,000 known drug-drug interaction pairs from DrugBank

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Models & Techniques

1. BioBERT (Mechanism Understanding)

- Pre-trained on biomedical literature
- Fine-tuned on drug mechanism descriptions
- Converts text → 768-dim embedding

2. Molecular VAE (Generation)

- Encoder: SMILES → latent space (256-dim)
- Decoder: latent space → valid SMILES
- Trained on 1M drug-like molecules from ZINC
- Generates structurally diverse candidates

3. Graph Neural Networks (Interaction Prediction)

- **Pairwise GNN:** 6-layer MPNN predicting CYP competition, receptor binding, toxic metabolites
- **Three-way GNN:** Combined model predicting emergent A+B+C effects
- Training: DrugBank DDI pairs (140K) + TWOSIDES adverse events (1M)
- Architecture: 256 hidden dims, 8 attention heads per layer

4. Safety Classifier

- Multi-task neural network predicting lethality, organ toxicity, cardiac risk
- Training: FDA adverse event reports, Tox21 toxicity data
- Output: Risk scores (0-10) for each safety dimension

Technology Stack: PyTorch, PyTorch Geometric, RDKit, Transformers (HuggingFace), DeepChem, Streamlit

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10-Week Timeline

Week	Tasks	Deliverable

1-2	Environment setup, download all databases, preprocess ChEMBL/DrugBank/TWOSIDES, build data pipelines, implement BioBERT mechanism interpreter	Can convert mechanism text → embeddings, 50K molecules processed into graphs
3	Implement molecular VAE, train on ZINC drug-like subset, validate chemical validity, generate test variations	Can generate 50 valid molecules per mechanism
4	Build pairwise interaction GNN, train on DrugBank DDI data, implement CYP/receptor/metabolite predictors	Can predict if two known drugs interact (75%+ accuracy)
5	Extend to novel molecule pairs, train three-way interaction model, implement metabolite cascade modeling	Can predict interactions for new molecules
6	Build hierarchical filtering pipeline, implement GPU batch processing, optimize for 125K combinations	End-to-end pipeline: mechanism → 1000 ranked combinations
7	Implement safety filters, build multi-objective scoring, validate on 3 therapeutic areas (pain, muscle spasms, malaria)	System generates safe, ranked combinations
8	Create 10 validation cases (5 known combinations + 5 novel), document prediction accuracy, compare with literature	System validated on known drug combinations
9	Build Streamlit web interface (4 pages), add visualizations, deploy to cloud	Public web app functional
10	Documentation, technical report, presentation slides, demo video, final testing	Project complete, ready to present

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Example Output: 3-Compound Muscle Relaxant

USER INPUT: "Reduce muscle spasms through calcium channel blocking and GABA enhancement without excessive sedation"

PROCESSING: 20 minutes analyzing 125,000 combinations

TOP RESULT:

Compound A: L-type calcium channel blocker | IC50: 12 nM | Dose: 40mg
Compound B: α2/α3-selective GABA-A agonist | EC50: 45 nM | Dose: 25mg
Compound C: CYP3A4 metabolism enhancer | Dose: 15mg

Pairwise Safety:

- A ↔ B: Risk 1.8/10 (Different targets, synergistic 1.35×)
- A ↔ C: Risk 0.5/10 (C enhances A absorption +30%)
- B ↔ C: Risk 1.2/10 (No metabolic conflict)

Three-Way Analysis:

- Combined risk: 2.1/10 (SAFE)
- Net synergy: 1.52× vs monotherapy
- No toxic metabolites predicted
- All use different CYP enzymes (no competition)

Predicted Outcomes:

- Efficacy: 87% response rate
- Safety: 9.4/10
- Sedation: LOW (0.32 probability)
- 42% more effective than standard treatment

Warnings: ⚠️ May cause mild dizziness, avoid alcohol, contraindicated in pregnancy

Next Steps: Synthesize compounds → Test in vitro → Validate interactions → Animal studies

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Computational Strategy

Challenge: 125,000 combinations is computationally expensive

Solution: Hierarchical Filtering

Stage 1 (GPU: 2 min): Quick pre-screening eliminates obviously bad combinations
Stage 2 (GPU: 5 min): Pairwise GNN checks (3 models run in parallel)
Stage 3 (GPU: 8 min): Three-way GNN on promising candidates
Stage 4 (CPU: 3 min): Detailed scoring and ranking
Stage 5 (CPU: 1 min): Generate explanations for top 10

Total time per query: ~20 minutes (acceptable for research tool)

Hardware: Google Colab Pro Plus (A100 GPU, 40GB VRAM) processes 1,000 combinations in parallel

Optimization techniques:

- Early stopping: Don't fully analyze combinations that fail safety checks
- Result caching: Store pairwise results, reuse for three-way
- GPU batching: Process 1,000 molecules simultaneously
- Lazy evaluation: Only compute detailed analysis for top 1,000

Performance Targets & Validation

Metric	Target	Validation Method
Mechanism understanding	80%+ accuracy	Manual review of 100 interpretations
Molecule validity	95%+	RDKit chemical sanitization
Pairwise DDI prediction	75%+ ROC-AUC	DrugBank held-out test set
Three-way prediction	65%+	Limited validation on known triplets
Safety filter recall	95%+	Must catch all known lethal pairs
Processing speed	<30 min	User experience benchmark
Top 10 quality	Expert approval	HCC chemistry professor review

Validation Strategy:

1. Test on 5 known 3-drug combinations from literature (should rank in top 100)
2. Generate 5 novel combinations, compare predictions with chemical intuition
3. Submit top 10 for each therapeutic area to chemistry faculty for review
4. Document where predictions differ from literature and investigate why

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Budget & Resources

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Total Cost: $192

- Google Colab Pro Plus: $50/month × 3 months = $150

- Domain (compoundiq.com): $12
- Backup compute/storage: $30
- All data: FREE (public databases)
- All software: FREE (open-source)

Alternative: If HCC has GPU cluster access, cost reduces to $12 (domain only)

Time Investment: 150-200 hours over 10 weeks (15-20 hrs/week)

Deliverables

1. **Functional Web Application**
    - Live at compoundiq.streamlit.app
    - 4-page interface: Input → Top 10 detailed → Top 500 table → Download 1000
    - Interactive visualizations, sortable tables, downloadable datasets
2. **Trained Models**
    - BioBERT for mechanism interpretation
    - Molecular VAE for generation (50 candidates per position)
    - 3 pairwise GNN models for interactions
    - 1 three-way GNN model for combined effects
    - Multi-task safety classifier
3. **Validation Results**
    - 10 test cases documented with predictions vs. literature
    - Performance benchmarks on all metrics
    - Expert review feedback from chemistry faculty
4. **Complete Codebase**
    - GitHub repository: github.com/[username]/compoundiq
    - Jupyter notebooks: data processing, model training, validation
    - Documentation: README, API reference, user guide
    - Installation instructions
5. **Presentation Materials**
    - 20-slide PowerPoint with live demo
    - 7-minute demonstration video
    - Technical report (10-12 pages)
    - Poster for showcase (optional)

Why This Matters

For Pharmaceutical Research:

- Current process: Scientists manually design combinations, test hundreds in lab, takes years
- With CompoundIQ: AI suggests top 1,000 candidates in 20 minutes, scientists test only promising ones
- Potential impact: 80% reduction in early-stage screening time, discover non-obvious synergies

For AI Education:

- Demonstrates deep learning on complex, multi-modal data (text + graphs)
- Shows real-world application with measurable impact
- Combines NLP, generative models, GNNs, multi-task learning
- Addresses safety-critical AI (wrong predictions could harm patients)

For Community Colleges:

- Makes advanced computational chemistry accessible
- Free alternative to commercial platforms ($100K+ licensing)
- Teaching tool for future molecular ML courses
- Foundation for student research projects

For Career:

- Publication-worthy (NeurIPS AI for Science workshop, ICML)
- Portfolio piece for pharmaceutical AI positions
- Demonstrates end-to-end system development
- Shows ability to handle computational complexity

Risks & Mitigation

Risk	Impact	Probability	Mitigation
125K combinations too slow	High	Medium	Hierarchical filtering, GPU optimization, fallback to 2 compounds
Three-way interactions hard to validate	Medium	High	Focus validation on pairwise, use literature for qualitative check
Generated molecules invalid	High	Low	Strict chemical validity checks, train VAE on drug-like molecules only
Interaction predictions inaccurate	High	Medium	Ensemble models, conservative safety thresholds, expert review

Web app deployment issues	Medium	Low	Streamlit Cloud proven platform, have local backup

Fallback Plan (if behind schedule by Week 6):

- Reduce to 2 compounds (50×50 = 2,500 combinations, much faster)
- Reduce candidates per position (50 → 30 per compound)
- Show top 100 instead of 1,000
- All changes can be made in 1-2 days

Success Criteria

Technical Success:

- ✓ System processes 125K combinations in <30 minutes
- ✓ Finds >1,000 safe combinations for each mechanism
- ✓ Pairwise predictions match known DDIs (75%+ accuracy)
- ✓ Top 10 combinations pass expert chemistry review
- ✓ All generated molecules are chemically valid (95%+)

Deliverable Success:

- ✓ Web app deployed and functional
- ✓ All 5 trained models working
- ✓ Complete documentation on GitHub
- ✓ 10 validation cases documented
- ✓ Presentation ready for defense

Educational Success:

- ✓ Demonstrates advanced deep learning techniques
- ✓ Shows real-world problem solving
- ✓ Creates reusable educational resource
- ✓ Contributes to HCC AI program reputation

Future Extensions (Post-Capstone)

Phase 2 (If continued):

- Expand to 4-5 compound combinations

- Add dosage optimization (find optimal mg for each compound)
- Implement pharmacokinetic modeling (blood concentration over time)
- Add synthesis route planning (how to make compounds in lab)
- Partner with pharmaceutical company for validation
- Publish in peer-reviewed journal
- Patent the system architecture

Research Questions to Explore:

- Can AI discover novel synergies humans miss?
- How accurate are three-way interaction predictions?
- What's the optimal number of compounds per formulation?
- Can this reduce animal testing requirements?

Conclusion

CompoundIQ represents an ambitious but achievable capstone project that pushes the boundaries of what's typically expected at the community college level. By designing 3-compound formulations with comprehensive interaction modeling, the system addresses a real pharmaceutical need while demonstrating advanced AI techniques including NLP, generative models, graph neural networks, and multi-objective optimization.

The 10-week timeline is tight but realistic with hierarchical filtering and GPU optimization. The deliverable—a functional web application that generates 1,000 ranked drug combinations in 20 minutes—would be impressive for any AI program and provides a foundation for future research or commercial development.

This project is ready for approval and implementation.

🔎 Feature Addition: User-Specified Compound Testing

CompoundIQ will include a Testbench Mode that allows users to evaluate specific named compounds instead of only generating new ones

CompoundIQ_ AI-Driven Drug Comb…

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How It Works

Users can input:

- **3 compounds → Full safety and interaction evaluation**
- **2 compounds + mechanism → System proposes optimal third compound**
- **1 compound + mechanism → System proposes two compatible partners**

A new Compound Resolver module will map drug names to canonical IDs and molecular structures before running the existing interaction prediction and safety pipeline

CompoundIQ_ AI-Driven Drug Comb…

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Output

The system will return:

- **SAFE / CAUTION / REJECT status**
- **Pairwise and three-way risk scores**
- **Predicted synergy**
- **Safety explanations (e.g., CYP conflicts, QT risk, toxic metabolite probability)**

Impact

This feature expands CompoundIQ from a generative design tool into a practical decision-support system for testing clinician- or literature-selected combinations